BING-YUE WU

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EDUCATION

Arizona State University 2023 – Present
Doctoral Studies in Electrical Engineering *Tempe, Arizona*
- Ph.D. candidacy achieved, 2026

National Taiwan University of Science and Technology (Taiwan Tech, NTUST) 2021 – 2023
M.S. in Electrical Engineering *Taipei, Taiwan*

National Taiwan University of Science and Technology (Taiwan Tech, NTUST) 2016 – 2020
B.S. with major in Electrical Engineering and minor in Computer Science and Information Engineering *Taipei, Taiwan*

SKILLS

Programming Languages: C, C++, Python, Tcl
EDA Tools: OpenROAD, OpenSTA, Innovus, Genus, ICC2, Design Compiler, HSpice, Virtuoso, Calibre
AI Tools: Claude Code CLI, Codex CLI, Cursor CLI
Language Ability: Mandarin (Native), English (Fluent)

PUBLICATIONS

B.-Y. Wu, C.-T. Ho, H. Yang, B. Khailany, and V. A. Chhabria, "**EvoDRC: A Self-Evolving Agentic Framework for Automated DRC Violation Repair**", *in Proc. ASP-DAC*, 2027. [GitHub]

V. A. Chhabria, A. Dey, A. B. Kahng, I. Savidis, P. Shrestha, and **B.-Y. Wu**, "**ResizerAgent: Agentic Strategy Selection for Timing Optimization in OpenROAD Resizer**", *in Proc. MLCAD*, 2026. [GitHub]

Z. Jiang, Q. Wu, **B.-Y. Wu**, and J. Zhang, "**CAPO: Certification-Guided Agentic Workflow for Physical Design Parameter Optimization**", *in Proc. GLSVLSI*, 2026.

B.-Y. Wu, C.-T. Ho, H. Yang, C.-C. Chang, A. B. Akkur, B. Khailany, and V. A. Chhabria, "**Special Research Session: Toward Agentic Solution for DRC Challenges in Digital VLSI Design**", *in Proc. DAC*, 2026. [GitHub]

B.-Y. Wu, A. Dey, A. Rovinski, and V. A. Chhabria, "**Focus Session: Large Language Models in Physical Design: From Data Generation to Intelligent Agents**", *in Proc. DATE*, 2026.

B.-Y. Wu and V. A. Chhabria, "**DALI-PD: Diffusion-based Synthetic Layout Heatmap Generation for ML in Physical Design**", *in Proc. ASP-DAC*, 2026. [GitHub]

V. A. Chhabria, A. Ghose, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Y. Liu, Z. Wang, and **B.-Y. Wu**, "**IEEE DATC RDF-2025: Enabling an EDA Research Ecosystem**", *in Proc. ICCAD*, 2025.

B.-Y. Wu, U. Sharma, A. Rovinski, and V. A. Chhabria, "**OpenROAD Agent: An Intelligent Self-Correcting Script Generator for OpenROAD**", *in Proc. ICLAD*, 2025. [GitHub]

V. A. Chhabria, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Z. Wang, **B.-Y. Wu**, and D. Yoon, "**Strengthening the Foundations of IC Physical Design and ML EDA Research**", *in Proc. ICCAD*, 2024.

V. A. Chhabria, **B.-Y. Wu**, U. Sharma, K. Kunal, A. Rovinski, and S. S. Sapatnekar, "**Generative Methods in EDA: Innovations in Dataset Generation and EDA Tool Assistants**", *in Proc. ICCAD*, 2024.

B.-Y. Wu, R. Liang, G. Pradipta, A. Agnesina, H. Ren, and V. A. Chhabria, "**2024 ICCAD CAD Contest Problem C: Scalable Logic Gate Sizing Using ML Techniques and GPU Acceleration**", *in Proc. ICCAD*, 2024. [GitHub]

U. Sharma*, **B.-Y. Wu***, S. R. D. Kankipati, V. A. Chhabria, and A. Rovinski, "**OpenROAD-Assistant: An Open-Source Large Language Model for Physical Design Tasks**", *in Proc. MLCAD*, 2024. [GitHub]

V. Gopalakrishnan, **B.-Y. Wu**, and V. A. Chhabria, "**ML-INSIGHT: Machine Learning for Inrush Current Prediction and Power Switch Network Improvement**", *in Proc. ISLPED*, 2024.

B.-Y. Wu, U. Sharma, S. R. D. Kankipati, A. Yadav, B. K. George, S. R. Guntupalli, A. Rovinski, and V. A. Chhabria, "**EDA Corpus: A Large Language Model Dataset for Enhanced Interaction with OpenROAD**", *in Proc. LAD*, 2024. **(Best Paper Nominated)** [GitHub]

V. A. Chhabria, W. Jiang, A. B. Kahng, R. Liang, H. Ren, S. S. Sapatnekar, and **B.-Y. Wu***, "**OpenROAD and CircuitOps: Infrastructure for ML EDA Research and Education**", *in Proc. VTS*, 2024. **(Primary Author)** [GitHub]

B.-Y. Wu, S.-Y. Fang, H.-W. Chang, and P. Wei, "**SpeedER: A Supervised Encoder-Decoder Driven Engine for Effective Resistance Estimation of Power Delivery Networks**", *in Proc. MLCAD*, 2022. **(Best Paper Award)**

Work Experience

Tenstorrent Sept. 2026 – Present
Advanced Physical Design Intern *Austin, Texas*

Arizona State University Aug. 2023 – Aug. 2026
Graduate Research Assistant *Tempe, Arizona*

- Researched LLM-based approaches for automated design rule violation (DRV) fixing and built a benchmarking framework to evaluate LLM-driven DRC methodologies.
- Proposed a diffusion-based generative framework to generate synthetic circuit layout heatmaps and open-sourced 23K+ synthetic samples to address data scarcity in ML-driven physical design research.
- Developed a large-scale OpenROAD LLM dataset, a domain-specific LLM for script generation and Q&A, and a self-correcting agent that iteratively refines scripts using tool feedback.
- Served as Topic Chair for the ICCAD 2024 CAD Contest (Problem C) and contributed to OpenROAD and the NVIDIA CircuitOps infrastructure to advance ML-EDA research and education.
- Hosted a tutorial at ASP-DAC 2024 on CircuitOps and the OpenROAD Python API, demonstrating Python-based physical design analysis and automation workflows for ML-EDA research.

Synopsys Inc. Oct. 2021 – June 2022
Intern (Technical-Engineering) *Taipei, Taiwan*

- Researched a novel Machine Learning-based (ML) solution estimating the effective resistance of Power Delivery Networks in advanced VLSI designs to speed up runtime and improve the accuracy of ML-driven IR analysis tools.
- Responsible for designing the data pipeline, the ML model architecture, and the entire effective resistance estimation workflow.

Academia Sinica Jul. 2020 – Nov. 2020
Full-time Research Assistant at Computational Finance and Data Analytics Lab *Taipei, Taiwan*

- Conducted research on a novel Transformer Encoder-based model with financial number category awareness, designing new pre-training and fine-tuning tasks to enhance its performance.
- Developed an Online Loan Application Recommender System for E.SUN Commercial Bank, achieving nearly 300% performance improvement. Responsibilities included workflow design, model development, and building Python APIs for industrial deployment.

Awards

Qualcomm Innovation Fellowship North America (QIF) May 2026
MLCAD Student Travel Grant Sept. 2024
Ferdinand A. Stanchi Fellowship Aug. 2024
DAC Young Fellow Travel Grant Jun. 2024
MLCAD Student Travel Grant Sept. 2023
Fulton Fellows Fellowship Aug. 2023
Best Paper Award at MLCAD 2022 Sept. 2022

Professional Service

Reviewer Activities
ACM TODAES (2025×2), IEEE TCAD (2026×3)

Topic Chair of Problem C at 2024 ICCAD CAD Contest Oct. 2024
IEEE CEDA *Newark, New Jersey*

2024 ASP-DAC Tutorial Talk Jan. 2024
ACM SIGDA *Incheon, South Korea*